CHAPTER 32

A Fundamental Role for Conceptual Processing in Emotion

Christine D. Wilson-Mendenhall and Lawrence W. Barsalou

The idea that there are consistent and discriminable patterns for a small number of "basic" emotions (e.g., fear, anger, sad, happy, disgust) motivates a great deal of emotion research (see Tracy & Randles, 2011, for a recent review of these models). This view is also pervasive outside of the scientific arena, in education, government policy, industry, and entertainment. The logic is clear: If we could discover these emotion-specific patterns, we would unlock the mystery of human emotions, because these basic emotions are also the foundation of more complex emotions. As science has a tendency to do, however, the empirical evidence is forcing us to pause. A more complicated picture is emerging, one in which there is tremendous variability in the physiological, neural, behavioral, and subjective changes that occur during emotions like fear, anger, happiness, sadness, and disgust (Barrett, 2006; Barrett et al., 2007; Lindquist, Siegel, Quigley, & Barrett, 2013; Lindquist, Wager, Kober, Bliss-Moreau, & Barrett, 2012; Ortony & Turner, 1990; Quigley & Barrett, 2014). Moreover, there are a great number of emotional experiences that do not fall into these categories. Perhaps, although elusive, some set of basic emotions is still the answer—the key to understanding this variability—and we should keep searching for them. But what if we could explain this variability without appealing to basic emotions? We suggest that conceptual processing offers a natural explanation of the variability in emotional life. Furthermore, conceptual processing offers a very different perspective on the functions of emotions than has traditionally been discussed in theories of emotion, and motivates a very different approach to studying them.

In this chapter, we explore possible roles that conceptual processing plays in emotion, drawing on both the cognitive and affective science literatures. In short, we suggest that concepts develop in memory for emotional experiences as they do for many other forms of experiences, shaped in large part by attention and learning and by the "niche" in which an individual operates, including internal bodily, external sensory, and sociocultural environments (Barrett, Wilson-Mendenhall, & Barsalou, 2015; Barsalou, 2009; Wilson-Mendenhall, Barrett, Simmons, & Barsalou, 2011). The emotion literature often emphasizes the role of conceptual processing and language in explicitly remembering and communicating emotional experiences. We propose that conceptual processing also plays a role in the implicit generation of emotional experiences. Concepts develop so that previous experiences can be used to guide responding in the present situation (via prediction and inference; Barsalou, 2003b, 2009; Barsalou, Niedenthal, Barbey, & Ruppert, 2003). Our goal for this chapter is to illustrate how conceptual processing could play a fundamental role in emotion.

Concepts Develop for Prediction and Inference

Concepts work so seamlessly most of the time that we are not aware that they are functioning. Consider a typical start to a weekday morning: you hear your alarm clock sound. This may seem like a simple event, but let us break it down. In more simple terms, you hear a sound that arouses your nervous system somewhat automatically due to its acoustical properties.[1] Now, what do you do? If you did not know anything about alarm clocks, you might begin by orienting toward the sound and trying to find its source. You do not do this, however. Instead, prior experiences with alarm clocks that exist in memory, which are organized in your concept of alarm clock, shape how you respond to this sound. Information not only about the alarm clock object itself (e.g., its visual features, sounds it produces, how you act on it, co-occurring changes in your internal state), but also information contained in other related concepts that underlie the situations that involve alarm clocks (concepts like morning, bed, sleep, refreshed, exhausted, etc.) guide how you respond to this sound. You know that you can control the sound (stop or delay it) and that it is a signal to wake up. Unlike a rigid conditioned response, you can act flexibly based on the situational context, because concepts underlying the larger situation are also available in memory and are integrated with your concept of an alarm clock. If exhausted upon hearing the alarm, you press the snooze button; if well rested upon hearing the alarm, you press the off button. Perhaps you set the alarm extra early the night before, so you hit the snooze button twice. Perhaps you forgot to set the alarm, and wake up on your own instead of hearing an alarm, so you do not hit any button, and instead search for the time display. Thus, a concept like alarm clock refers to a set of patterns in memory that share various similarities and can flexibly be used to guide action. It refers to the aggregation of information about category instances into some kind of integrated representation (Barsalou, 2003a, 2005, 2012; Barsalou & Hale, 1993; Murphy, 2002). Concepts underlie our ability to interpret and engage in the situations that characterize everyday life.

The purpose of concepts, in other words, is prediction—going beyond the information that is present to *infer* what will happen next and to shift the biological system in ways appropriate to the situation (in the form of motor actions, perceptual sampling, attention shifts, visceromotor regulation, metabolic changes, etc.; Barsalou, 2009). In this way, concepts actively *interpret and construe* experience, dynamically shaping moment-to-moment "responding" in the form of perceiving, coordinating action, regulating the body, and organizing thoughts. Many theorists emphasize the value of prediction (and inference) in terms of filtering sensory information from the external environment, which allows us to act quickly and efficiently, and to allocate appropriate resources to novel stimuli (e.g., Clark, 2013). Prediction is also valuable in terms of filtering sensory information from the body and initiating efficient visceromotor responding (e.g., Barrett & Bar, 2009; Barrett & Simmons, 2015; Seth & Critchley, 2013). Prediction also plays central roles in internally oriented thought and reasoning, filtering and allocating attention to organize or redirect thoughts, plan for the future, generate new ideas, and so on (Barsalou, 2009).

Concepts develop to interpret virtually every aspect of the situations we experience—externally oriented experiences of objects and scenes in the world (e.g., violin, banana, mountain, statue), internally oriented experiences of our bodies (e.g., hunger, pain, itch, sweet), complex mental experiences that often involve how our internal and external worlds meet (e.g., anger, doubt, belief, analyze), and social experiences and institutions that involve many minds and bodies (e.g., government, protest, religion, concert). Simple and concrete concepts typically emerge earlier in the lifespan,[2] with more complex and abstract concepts, including many emotion concepts, developing later (Bretherton & Beeghly, 1982; Bretherton, Fritz, Zahn-Waxler, & Ridgeway, 1986; Gilhooly & Gilhooly, 1980; Kuperman, Stadthagen-Gonzalez, & Brysbaert, 2012; Ridgeway, Waters, & Kuczaj, 1985; Wellman, Harris, Banerjee, & Sinclair, 1995). As we will see, many different situational patterns accrue for each concept so prediction occurs as efficiently as possible at any given moment, in any given situation—guiding perception, action, regulation, and decision making. Furthermore, concepts are dynamic and flexible. Attention and learning support accruing many different patterns for a concept, and continually updating and expanding these patterns. Sometimes the changes in the patterns that underlie a concept are dramatic, as we see for emotion concepts later.

In the sections that follow, we focus on the characteristics of conceptual processing that are most relevant for understanding how concepts underlie emotional experiences. We propose that concepts develop to efficiently mobilize changes that characterize affective and emotional experiences.[3] Learning situated conceptualizations can explain the tremendous variability in emotional experiences within an individual and across different individuals.

Attention Is Integral to Establishing Concepts

A concept reflects a coherent aspect of experience that is represented by a set of patterns in memory. Attention is central to acquiring (and using) concepts because the brain's attention systems select specific aspects of the current experience, which often become integrated with similar schematic experiences in memory (Barsalou, 1999). In contrast to a "snapshot" or "recording" of experience like what a camera would produce, specific concepts develop from extracting aspects of experience than can be used to interpret similar experiences in the future (e.g., red, apple, hungry; Barsalou, 1999; Schyns, Goldstone, & Thibaut, 1998). Parsing experience into concepts is productive because, later, this enables constructing an infinite number of possible situated experiences from a finite number of concepts. Because no two situations are exactly the same, it is critical for the brain to organize information in memory so it is readily available for interpreting the situation at hand, which often involves the dynamic assembly of concepts. In this section, we discuss how attention operates in ways that are integral to establishing and using concepts to navigate situations.

Selecting Meaningful Units

Selection is a core function of attention and reflects efficient processing of what is most relevant to ongoing goals and behaviors (Chun, Golomb, & Turk-Browne, 2011; Johnson & Dark, 1986; Peelen & Kastner, 2014; Treisman, 1969). Selection can operate on properties of one modality (e.g., visual, auditory, gustatory) and on dimensions of space, time, or intensity that typically occur across modalities (Bahrick, Lickliter, & Flom, 2004). We assume that selective attention operates on interoceptive sensorimotor activity grounded in the body in a parallel way to how it operates on externally oriented sensorimotor activity.[4]

Selective attention establishes the meaningful units upon which concepts are built. For example, the spread of attention across multiple sensory modalities while allocated to a specific location in space serves to bind local sensory information into a multisensory object concept (Busse, Roberts, Crist, Weissman, & Woldorff, 2005; Treisman, 1998). Initially, intersensory redundancy appears to drive selective attention (e.g., temporal synchrony, rhythm, tempo, and changing intensity that a person's face and voice often express; Bahrick et al., 2004). As attentional skills develop, coordinated attention across modalities guided by time, space, and intensity dimensions is central to learning the relations (e.g., above, after, cause, intend) that are critical components of many concepts (e.g., Cohen & Amsel, 1998; Cohen, Chaput, & Cashon, 2002; Logan, 1994). Although people are sometimes aware of attending to a specific aspect of experience (e.g., when studying, when meditating), selection processes operate continually outside of awareness, which underlies accruing and integrating the many patterns underlying concepts in memory.[5]

Top-Down Processing Becomes Routine

As concepts develop in memory, top-down conceptual patterns and bottom-up sensory input often interact to direct selection processes (Chun et al., 2011; Corbetta, Patel, & Shulman, 2008) that include orienting, filtering, searching, and expecting (Plude, Enns, & Brodeur, 1994). This interaction supports the transition from the rigid stimulus–response responding present in early life to more adaptive, situated responding. The mechanisms underlying stimulus-driven attention are important because they bias attention toward novel features of experience (e.g., Klein, 2000) and disrupt top-down cognitive control to reorient attention. These kinds of disruptions tend to occur, for example, with a temporally abrupt onset, change in intensity, appearance of new objects, and moving or looming stimuli (especially as an interaction becomes more probable) (Chun et al., 2011). Evidence increasingly suggests, however, that bottom-up, sensory-driven processing occurs relatively infrequently, reflecting an interruption in routine top-down processing when it does occur (Clark, 2013; Vetter & Newen, 2014). Whether a stimulus is sufficiently novel to reorient attention is dependent on the concepts accessible in memory.

Top-Down Processing Interprets and Construes

Through attention operations, top-down conceptual processing dynamically shapes moment-to-moment perception, action, regulation, and decision making (Barrett & Bar, 2009; Barsalou, 2009). Extensive evidence in perception, for example, demonstrates that top-down conceptual predictions change and distort sensory information based on what is predicted to occur, shaping attention and perception through expectation (Barsalou, 2009; Vetter & Newen, 2014). The brain coordinates mental activity through internally driven interpretation and prediction, not through reflexive, externally driven responding (e.g., Raichle, 2010). Computationally, in predictive coding neuroscience models, information from memory does not merely modulate incoming sensory activity, it drives it, which means it can change how sensory information is being sampled (Bastos et al., 2012). It is more efficient to change perceptions of the environment to be consistent with patterns in memory (i.e., what a person already implicitly knows).[6]

Because it is unclear whether veridical perception even exists, recent proposals suggest that discussing levels of conceptual "penetration" or "construal" may be more productive (Vetter & Newen, 2014). The predictions driving construal penetrate very early perceptual processing (McCauley & Henrich, 2006). Top-down color knowledge, for example, can be decoded from patterns in primary visual cortex (V1) when participants view achromatic objects (Bannert & Bartels, 2013). The top-down penetration of early sensory processing is not specific to vision, occurring in other sensory modalities and during multisensory integration (Barsalou, 2009; Vetter & Newen, 2014). Initial evidence suggests that top-down processing of interoceptive activity also occurs (Barrett & Bar, 2009; Barrett & Simmons, 2015; Petersen, Schroijen, Molders, Zenker, & Van den Bergh, 2014; Seth & Critchley, 2013). In our view, then, attention and other systems are typically operating via conceptual processing, with predictions that distort perception, facilitate motor actions, regulate the viscera (and so on) continually occurring, and with updating occurring in novel situations. As we see next, top-down conceptual processing is what underlies efficiently engaging in situations, including the situations in which emotions emerge.

Learning Establishes Flexible, Situated Concepts

Learning grounded in attention establishes concepts that support interpreting and navigating experience in increasingly complex ways. Learning a concept involves accruing instances—patterns in memory—for an aspect of an experience (Allen & Brooks, 1991; Barsalou, 1987; Medin & Schaffer, 1978; Murphy, 2002). As we will see, the patterns that become organized in memory for a concept, the different instances, reflect the situations in which they occur (Barsalou, 2003b, 2008b; Barsalou, Breazeal, & Smith, 2007; Yeh & Barsalou, 2006). We refer to representing a concept in a situation, a situated instance, as a *situated conceptualization* (Barsalou, 2003b, 2009; Barsalou et al., 2003; Wilson-Mendenhall et al., 2011). Developing situated conceptualizations supports prediction and inference through efficiently assembling concepts to dynamically interpret and construe what is occurring based on prior experiences (Barsalou, 2009). More simple, concrete concepts that reflect *local* components of situations (e.g., apple conceptualizes part of an event in which a family bakes an apple pie together during a holiday) tend to develop as focal aspects of situated conceptualizations (Barsalou, 2003b, 2008b). As learning occurs in situations and attention operations become increasingly driven by top-down assemblies of concepts during situated conceptualization, more complex, abstract concepts emerge. These concepts, which typically integrate local components into *configural relational* structures (e.g., tradition integrates various parts of the event, including apples, baking, family, and holiday), tend to develop as entire situated conceptualizations (Barsalou, 1999; Barsalou & Weimer-Hastings, 2005; Wilson-Mendenhall, Simmons, Martin, & Barsalou, 2013). In this section, we build on our discussion of attention to examine how increasingly complex concepts, including the concepts underlying emotional experiences, are learned. We demonstrate how language underlies very different patterns, situated conceptualizations, becoming organized as instances of concepts. We first discuss our general theoretical approach and then apply it to emotional experiences.

Implicit Statistical Learning Is Occurring Early

Because aspects of situations tend to occur repeatedly, the brain organizes these regularities together as concepts in memory to facilitate interpreting similar situations in the future. Simple concepts can develop very early via implicit mechanisms in the brain that extract the organizational structural or "statistical regularity" across instances in the stream of situated activity filtered through attention (Aslin & Newport, 2012; Turk-Browne, 2012). Statistical learning is a powerful learning mechanism that is operating well before an infant is a year old (Kirkham, Slemmer, & Johnson, 2002; Saffran, Aslin, & Newport, 1996). Initially, the system easily extracts physical regularities that tend to be stable in the external environment (e.g., the boundary between earth and sky is consistently horizontal, not vertical), perhaps due to evolved biological scaffolding that needs minimal tuning (Turk-Browne, 2012). As selective attention becomes more sophisticated, statistical learning also operates on aspects of situations that tend to co-occur together and exhibit stable relations to one another (e.g., sky is above the horizon and ground is below it; Turk-Browne, 2012). The patterns underlying concepts at this point may be relatively rigid, with more flexibility introduced as language develops.

Words Support Flexible Learning in Different Situations

A word can further facilitate statistical learning through its redundancy with repeated sensorimotor patterns that are never exactly the same, strengthening the association (e.g., the word "apple" co-occurring with similar taste sensations and actions during eating; Smith, Suanda, & Yu, 2014; Smith & Yu, 2008; Yoshida & Smith, 2005). The redundancy provided by the word facilitates situated conceptualization. When the word "apple" is used in future, similar situations, for example, the brain mobilizes patterns in memory, generating top-down predictions that shape taste perception and guide actions (Barsalou, 2009). A word also provides a redundancy that can guide increasingly dissimilar experiential regularities—extracted via attention—to become established for a concept. Experiences of apple during a grocery shopping situation, for example, differ from experiences of apple during an afternoon snacking situation (e.g., involving searching for apples in produce bins, placing them in a cart, paying for them, and so forth in the shopping situation vs. involving grasping apple slices, inserting them in the mouth, tasting and chewing them, and so forth in the snacking situation). Whereas the patterns that accrue upon using the word "apple" when eating a snack support interpreting and engaging in future snacking situations, the patterns that accrue upon using the word "apple" when grocery shopping support interpreting and engaging in future shopping situations. Different perceptions, actions, regulation, and so on emerge in the two situations (Barsalou, 2003b, 2008b). As we return to later, regularities that are established across various instances of a concept using language and attention, and that are most useful for top-down prediction, are often driven by goal-oriented action in situations (e.g., eating apples as a healthy snack, selecting apples in the grocery store, picking apples as a leisure activity), which become richer and more varied with increasing cognitive capacities and novel experiences (Barsalou, 1991, 2003b). Throughout life and especially during childhood, implicit acquisition of these regularities is often structured through interactions with others (e.g., parents, siblings, caregivers, teachers; Akhtar & Tomasello, 2000; Tomasello, 1992).

Initially, words are important for establishing the patterns that underlie local aspects of situations—the more simple, concrete concepts upon which complex, abstract concepts are built. Because learning operates as continuous situated activity cascades across the brain,[7] the patterns that become organized in memory for a concept, the different instances, are each integrated within a situation (Barsalou, 2003b, 2008b; Barsalou et al., 2007; Yeh & Barsalou, 2006). Some instances of apple, for example, might become integrated within a situated conceptualization involving the concepts cutting board, kitchen, and knife; whereas other instances of apple might become integrated within a situated conceptualization involving the concepts tree, orchard, and autumn; and still other instances within a situated conceptualization involving the concepts pie, oven, and grandma. Many empirical studies demonstrate the extensive presence of situational information as people develop and use concepts (e.g., Bar, 2004; Barsalou & Weimer-Hastings, 2005; Chaigneau, Barsalou, & Zamani, 2009; Wu & Barsalou, 2009; see Yeh & Barsalou, 2006, for a review). Rather than the concept functioning in a rigid manner across situations, it is functioning flexibly in widely varying sets of concepts that contextualize it in each situation.

Language Supports Learning More Complex Concepts

As concepts develop in situations, words transition from operating as a co-occurring redundancy for establishing patterns to operating as top-down mobilizers that efficiently construct increasingly elaborate situated conceptualizations (Barsalou, Santos, Simmons, & Wilson, 2008; Vigliocco, Meteyard, Andrews, & Kousta, 2009). Words become especially important for coordinating concepts such that they are integrated into the configural, relational patterns in situated conceptualizations that underlie many abstract and emotion concepts. When this type of learning is occurring, language—typically in the form of sentences, not single words—guides attention to various aspects of a situation (Zwaan & Madden, 2004; Zwaan & Radvansky, 1998). Language supports productively assembling and combining concepts to establish more complex concepts, often through syntax that facilitates recursive embedding, mental time travel, and other configural, relational operations (Barsalou, 1999, 2008a; Clark, 2006; Schmid, 2000). Because language draws on internally oriented attention processes associated with working memory to assemble more elaborate situated conceptualizations, abstract concepts typically emerge later in development.

Consider, for example, a situation in which an individual tells her grandchildren that baking apple pie at this time every year is a holiday tradition as she places the pie in the oven. Language that situates the word "tradition" guides the top-down coordination and integration of concepts into a situated conceptualization underlying the concept of tradition. Assembling concepts into a situated conceptualization, linked to a specific word (e.g., the word "tradition"), supports efficient interpretation and engagement in future, similar situations. As we will see, it coordinates trajectories of moment-to-moment prediction and inference that guide perceiving, coordinating action, regulating the body, and organizing thought. It also supports efficient communication and interaction with others (Barrett, 2012). Learning concepts in this way often remains implicit in the sense that individuals are not aware of having learned a concept—they did not intend to learn a concept or remember how they learned it.

As the word "tradition" and supporting language is used in different situations, different situated conceptualizations develop that underlie the concept (e.g., for holiday traditions involving annual activities, for religious traditions involving weekly rituals, for vacation traditions involving travel with friends; Barsalou, 2003b, 2008b). Because the situated conceptualizations reflecting different instances of an abstract concept can vary dramatically, goals often represent what is regular and co-occurring across subsets of them (e.g., goals like sharing an enjoyable activity with family, teaching an important skill to others, and performing a valued duty in the community, for the abstract concept of tradition).[8] Importantly, though, goal-driven regularities are not particularly useful for interpreting and engaging in the situation without the situation-specific assembly of concepts embedded in the situated conceptualization (Wilson-Mendenhall et al., 2011).

Learning Concepts for Emotional Experiences

We assume that many emotion concepts develop like abstract concepts do (see also Vigliocco et al., 2009), with situated conceptualizations that assemble concepts developing as an emotion word and supporting language is used in different situations. These situated conceptualizations emerge out of the more simple concepts that develop for aspects of affective experiences, including internal states. Through a discussion that parallels that above, we highlight empirical examples from across the lifespan to illustrate how situated conceptualizations underlying emotions could be learned.

Learning Statistical Regularities Early

We assume statistical regularities in the body's internal environment (e.g., underlying hunger, arousal) could be learned in an analogous manner to those in the external environment. We further assume that regular situational co-occurences involving interoceptive activity could be learned in this manner. The simple reflexes that newborns display in response to the distress of overwhelming arousal (e.g., eye closing, head aversion, non-nutritive sucking) disappear quickly as more adaptive learned patterns become established in memory from experience (Kopp, 1989).[9] Infants learn situational co-occurrences among their internal states, their caregivers' actions, and their own actions as caregivers play an increasingly central role in regulating infants' arousal states (Kopp, 1989), which support interpreting and actively engaging in future, similar situations. Different cries, for example, can be used to signal hunger, pain, or boredom to a caregiver. The parent–infant synchrony that presumably supports statistical learning during the first year predicts conceptually driven capacities for symbolic play and internal state talk at 2–3 years of age (Feldman, 2007; Feldman & Greenbaum, 1997).

By 8 months of age, infants are establishing conceptual patterns that allow them to anticipate the affective changes that result from others' actions (e.g., cringing in anticipation of the physical discomfort involved in a vaccination, becoming distressed when an attachment figure prepares to leave; Bretherton et al., 1986).[10] The concepts infants are using, although rudimentary, appear to be increasingly flexible and situated (i.e., flexibly drawing on different instances of a concept). For example, 11-month-old infants express distress and increase attention when mom disappears into a closet, relative to when mom disappears through a doorway, suggesting that they are sensitive to the situation in which separation occurs (Littenberg, Tulkin, & Kagan, 1971).

Learning in Situations Using Words

We assume that initial word learning facilitates accruing patterns that underlie internal state concepts and other concepts in affective situations—an important precursor to developing more complex emotion concepts. Infants understand and begin to refer to the basic physiological and affective internal states that are central to daily routines and situations (e.g., sleepy, hungry, good, happy, tired, hot) as they learn words throughout the second year (Benedict, 1979; Bretherton & Beeghly, 1982; Ridgeway et al., 1985). We assume that different instances of these concepts accrue as they are named in different situations, and that they are increasingly integrated with other concepts as situated conceptualizations develop. We further assume that the concepts that are developing in affective situations during this time tend to be those that are useful for interpreting and engaging in the situations (and are heavily dependent on interactions with caregivers). When infants express distress or delight, for example, caregiver language tends to center around the eliciting situation and how to concretely resolve/maintain the affective state instead of labeling the entire situation using an emotion word (like "fear," "anger," "sadness," "joy"; Bloom, 1993).

Only later, into their third year of life, do children begin to use words to name and understand discrete emotional experiences (e.g., "sad," "afraid," "scared," "angry," "mad"; Bretherton & Beeghly, 1982; Bretherton et al., 1986; Ridgeway et al., 1985; Wellman et al., 1995). Interestingly, the concepts that are developing in association with these words appear to initially be anchored on patterns of internal states underlying distress, paralleling the development of concrete, simpler concepts. Children often confuse emotions that feel unpleasant during this time (like anger and sadness), and it is not until later in the preschool years that children's emotion concepts differentiate emotions of the same valence effectively (Widen & Russell, 2008, 2010). We assume that as language (i.e., phrases and sentences) and working memory operations develop, situated conceptualizations are constructed that support differentiating affective experiences using concepts like fear, anger, sadness, and so on.

Learning Integrated Situated Conceptualizations

As language and attention skills develop in a communicative context that supports learning, we assume that integrated situated conceptualizations underlying emotion concepts are learned (Barrett, 2009, 2013; Lindquist, Satpute, & Gendron, 2015). Assembling a configuration of concepts into a situated conceptualization through language, and linking the situated conceptualization to a specific emotion word (e.g., "fear," "anger," "sad"), supports efficiently interpreting and engaging in future, similar situations. The developmental literature provides many examples of parent–child interactions that involve discussing an emotion in the context of specific situations, including the events that preceded an affective state, the expressive actions that accompany an affective state, and subsequent coping actions (Bretherton & Beeghly, 1982; Bretherton et al., 1986). Discussing a child's experience during a vaccination situation as involving fear (being scared, afraid), for example, might include talking about the restlessness and discomfort of anticipating the shot, the interactions with the doctor, the comforting presence of the parent (holding the child's hand, etc.), the needle instrument and the actions involved in the injection, the pain experienced and expressive crying, and so on. Integrating concepts underlying these aspects of the experience into a situated conceptualization underlying fear supports later mobilizing them to efficiently generate series of implicit, top-down predictions during future vaccination situations (which drive changes in perception, action, and regulation that we explore in the section "Mobilizing Coordinated Concepts in Situations"). The constructive nature of this process suggests that the situated conceptualizations underlying emotions like fear, anger, sadness, and so on may change quite dramatically in the early years, becoming increasingly elaborate to the extent that such language-based interactions occur.

Across the lifespan, we assume that many different situated conceptualizations are developing for a given emotion concept. Situated conceptualizations underlying fear, for example, might develop in situations that involve undergoing a painful medical procedure, interacting with an intimidating stranger, disappointing a love one, riding a rollercoaster, being rejected by one's peers, injuring oneself in a car crash, starting a challenging new job, damaging one's reputation, and so on.[11] Empirical work increasingly demonstrates that by adulthood, varied situated conceptualizations—instances—underlie any given emotion concept (Barrett, 2009; Lebois, Wilson-Mendenhall, Simmons, Barrett, & Barsalou, 2016; Oosterwijk, Mackey, Wilson-Mendenhall, Winkielman, & Paulus, 2015; Wilson-Mendenhall et al., 2011). Adults can describe different instances of emotions like anger from their own life (Russell & Fehr, 1994). Moreover, they indicate that some of these instances are more typical than others, when instances are presented as specific words (e.g., "fury" is a more typical example of anger than "impatience" or "discontent"; Fehr & Russell, 1984, 1991; Russell, 1991; Russell & Fehr, 1994) or as different situational scenarios (Condon, Wilson-Mendenhall, & Barrett, 2014; Wilson-Mendenhall, Barrett, & Barsalou, 2015). Behavioral and neuroscience evidence suggest that these within-concept typicality ratings are meaningful, with atypical category instances of emotion concepts like fear, anger, and sadness processed less efficiently than typical instances (Russell & Fehr, 1994; Wilson-Mendenhall et al., 2015). Several recent studies provide further evidence that instances of emotion concepts vary, demonstrating that different forms of the same emotion produce different neutral patterns (e.g., oral, bodily, and moral disgust; Harrison, Gray, Gianaros, & Critchley, 2010; Moll et al., 2005; von dem Hagen et al., 2009), context shapes the experience of an emotion (e.g., fear during physical danger differs from fear during social evaluation; Wilson-Mendenhall et al., 2011), and individuals experience the same emotional situation differently (e.g., experiences of anger are dependent on a person's interpretation of an unpleasant situation; Kuppens, Van Mechelen, Smits, De Boeck, & Ceulemans, 2007).

We assume that the various situated conceptualizations underlying an emotion concept may tend to cluster around typical goals or motivations (e.g., avoiding anticipated harm/distress for fear) or emergent feelings (e.g., feeling unpleasant for fear), but that it is unlikely that these regularities are always present. The fear involved in riding a rollercoaster, for example, likely involves different goals (e.g., experiencing a thrill) and may even involve pleasant feelings. Moreover, as described above, goal-driven regularities are not particularly useful for interpreting and engaging in a situation without the situation-specific assembly of concepts embedded in the situated conceptualization (Wilson-Mendenhall et al., 2011).

Mobilizing Coordinated Concepts in Situations

Accruing situated conceptualizations supports the top-down coordination of concepts to efficiently navigate and learn in the situation at hand. Patterns of situated activity are continually interpreted through trajectories of situated conceptualization as a situation is experienced (either because it is presently occurring, or because it is occurring through reconstructing the past or planning/imagining the future; Barsalou, 2003b, 2008b; Barsalou et al., 2007; Yeh & Barsalou, 2006). As this occurs, the brain goes beyond the information present to infer what will happen next and to shift the biological system in ways appropriate to the situation (in the form of motor actions, perceptual sampling, attention shifts, visceromotor regulation, metabolic changes, etc.; Barsalou, 2009). Building on the empirical evidence presented in previous sections, we now address how dynamic situated conceptualization occurs for increasingly complex concepts, concluding with emotion concepts. We approach this theoretical discussion from the perspective of a typically functioning adult mind that developed the capacity for increasingly complex attention and learning as described above. At this point, we assume top-down prediction and inference is routine because concepts accumulated to interpret repeated situations (Barsalou, 2009; Vetter & Newen, 2014) and that the brain's neural architecture is, at many levels, organized for prediction (Barrett & Simmons, 2015; Bastos et al., 2012; Friston, 2012; Friston & Kiebel, 2009; Kroes & Fernandez, 2012; Shipp, Adams, & Friston, 2013). Because no two situations are exactly the same, we further assume that situated conceptualization is a dynamic process that reflects probabilistic activity and tuning, and when we refer to situated conceptualizations *mobilizing*, we are referring to this active, probabilistic processing (Barsalou, 2011).

Mobilizing Local (Often Concrete) Concepts in Situations

In familiar situations, situated conceptualizations can mobilize quickly and fluidly without explicit direction from language (although word forms are likely activated implicitly via pattern completion as an associated part of the situated conceptualization). Consider a situation in which an individual is searching for a healthy snack in the kitchen, and sees an apple. In this situation, with goal-directed conceptual processing of kitchen, healthy, snack, and hungry already occurring, visual processing of the apple provides the critical constituent of the pattern needed to mobilize a situated conceptualization that involves eating the apple. The brain quickly goes beyond the visual experience of the apple to initiate changes in the brain and body. Pattern completion inferences provide educated guesses about what is likely to occur next—inferring actions to execute (e.g., grasping, biting, chewing), external and internal sensations that will occur (e.g., sweet taste sensations, crunchy sound sensations, satiating body sensations), internal regulation to initiate (e.g., metabolic activities), affective feelings that will emerge (e.g., the pleasant satisfaction of choosing a healthy snack), and so forth. The rapid unfolding of top-down predictions prepares and *engages* the body and mind by initiating action and regulation involved in eating the apple (Barrett & Bar, 2009; Papies & Barsalou, 2015). Top-down predictions drive attention and perceptual sampling in the situation, changing the way it is experienced. Because this particular situated conceptualization is driving perception, for example, the apple may appear a more appetizing shade of red than it would otherwise (e.g., if the situated conceptualization did not involve eating the apple).

In contrast, consider a situation in which an individual is searching for apples in the produce section of the grocery store. The situated conceptualization mobilizing in this situation generates a very different set of top-down predictions as the situation unfolds—predictions that drive perception and attention to visual features that distinguish apples from other fruits, initiate executing actions involved in selecting apples and placing them in a cart, and so on. Different situated conceptualizations involving a given concept are constructed to tailor prediction and inference to the situation at hand. The regularities imposed by situations are critical for using concepts to shift the brain and body into states that are consistent with an individual's goals and needs, which can vary dramatically from situation to situation (Barsalou, 2009).

For situations that are highly familiar, situated conceptualization reflects an assembly of concepts that mobilize as a trajectory of top-down predictions, which may continue for as long as patterns continue to loosely match perceptual input and reafferent signals, coordinating attention, perception, action, interoception, and so forth. When there is a considerable discrepancy between experience and the pattern mobilized in the situated conceptualization (i.e., what could be considered novelty), attention operations will interrupt the top-down flow of inferences that are occurring. At this point, situated conceptualization may change dramatically, reassembling concepts to reflect the multimodal change in experience or, alternatively, conceptual processing could become more fragmented as learning and updating occurs in specific modalities. Although the complexity and coordination of conceptual processing may change dynamically in this manner, the brain is always producing predictions at some level to interpret experience. Consider a situation in which an apple that an individual intended to eat feels mushy and smells odd. In this situation, the conceptual processing will begin to produce different predictions related to foods with unusual textures and smells, which might prepare the individual to dispose of the apple. Furthermore, because the salient sensations involved reorienting attention, further encoding and retrieval will likely follow, establishing a situated conceptualization involving rotting apples.

Mobilizing Configural (Often Abstract) Concepts in Situations

Situated conceptualization occurring at any given point in time constrains the situated conceptualization that follows. Consider a situation in which an individual's goal is to prepare for a large meal the morning of a holiday (e.g., American Thanksgiving). Visual processing of an apple mobilizes a situated conceptualization that involves preparing an apple pie. At this moment in time, predictions tailored to apple occur (e.g., motor actions like slicing, taste sensations like sweet) and coordinated concepts mobilize (e.g., knife, sugar, oven), guiding shifts in attention that accompany changes in perceiving, coordinating action, regulating the body, and organizing thoughts. Situated conceptualization dynamically changes as this occurs, generating new predictions from moment to moment to infer what to do next (e.g., to locate and grasp the knife, search the cabinet for sugar).

Through such trajectories, constraints accrue as goal-directed situated conceptualization guides prediction and inference involving local concepts in a situation, and as configural relations among these concepts are integrated in working memory. With specific situational constraints in place, a shift in the assembly of concepts underlying an individual's goals often appears to provide the critical constituent for mobilizing a situated conceptualization underlying a more complex, abstract concept. In the situation described above, for example, the individual's goal might shift from preparing for the meal to sharing an enjoyable activity with family when the kids wander into the kitchen. At this point, the concepts underlying the goal of sharing an enjoyable activity—configuring with concepts underlying family, apples, pie, baking, and holiday—provide the critical constituents for mobilizing a situated conceptualization underlying tradition.

Because a concept like tradition is more complex, the underlying situated conceptualization assembles concepts configured in time and space that generate a series of predictions (inferring what is going to occur). Situated conceptualization, as we refer to it here, really refers to extended trajectories of situated conceptualization that coordinate series of predictions. Linked local concepts not yet perceived mobilize via pattern completion, producing predictions that dynamically shape moment-to-moment experience in the form of perceiving, coordinating action, regulating the body, and so on as described above. Because this particular situated conceptualization is driving perception, for example, the individual might perceive family members' smiles as broader and more intensely positive than they would otherwise, consistent with the goal of sharing an enjoyable activity. Goals often shape the local concepts that are assembling in such situated conceptualizations. If an individual's goal is to share an enjoyable experience, for example, assemblies of local concepts will organize around actions shaping interactions between family members. On the other hand, if an individual's goal is to teach his or her children to execute a family recipe, the assemblies of local concepts will organize around actions shaping precise pie baking.

Similar to other concepts, we assume many different situated conceptualizations, grounded in goals, are constructed for complex, abstract concepts to tailor prediction and inference to the situation at hand. Situated conceptualizations underlying the concept of tradition could include holiday traditions involving annual activities, religious traditions involving weekly rituals, vacation traditions involving travel when friends are available, and so on. We further assume that the situated conceptualizations underlying abstract concepts are largely assembled based on prior learning, with learning and updating occurring in novel situations as described above. Because language is often involved in these more complex experiences, and because abstract concepts involve assemblies of concepts, "ad hoc" variations of situated conceptualizations underlying an abstract concept may occur relatively frequently.

Mobilizing Emotion Concepts in Situations

Initiating shifts in the brain and body quickly is often especially important during emotional experiences. Situated conceptualizations serve this function by coordinating concepts to prepare and engage the body as the situation unfolds. Because the concepts underlying emotional experience often develop as abstract concepts (i.e., that integrate local concepts into configural relational structures), trajectories of situated conceptualization can be initiated when they mobilize. Pattern completion inferences across space and time generate predictions and initiate changes quickly and fluidly.

Consider a situation in which an individual is driving home late in the evening, after indulging in a delicious holiday meal and drinking several glasses of wine. Resting her eyes for just a moment, the car starts to drift off the road. When her hands slip from the steering wheel, she jerks awake. Through a rapid trajectory of situated conceptualizations, key constraints of the situation then emerge (i.e., actions involved in awaking, perceptions of the car drifting off the road) that shift her goal to avoiding the potential harm involved in a car crash. The assembly of these concepts mobilizes a situated conceptualization underlying fear. Linked local concepts not yet perceived mobilize via pattern completion to initiate specific changes in the brain and body (in the form of perceiving, coordinating action, regulating the body, organizing thoughts). For this situation, in which linked local concepts underlying driving are central, visceromotor changes might occur that underlie scanning the road for other cars and activating muscle tension to grip the steering wheel. Shifts in attention and perceptual changes occur such that surrounding cars appear closer and sound louder than they would otherwise. Motor actions occur that support turning the steering wheel and decelerating such that the car stays on the road. Thus, the situation is experienced as "threatening" because the brain is constructing situated conceptualizations to interpret and engage in the situation.

If the concepts assembled in the situated conceptualization were not available in memory (including the concepts underlying car, steering wheel, driving, crashing, etc.), the situation would not be so readily perceived as threatening. Furthermore, the changes that occur during a situation like this one will differ across individuals because situated conceptualization reflects prior learning (i.e., instead of changes reflecting stereotyped patterns of responding that are consistent across individuals).

The changes that occur during situated conceptualizations underlying an emotion concept vary greatly from situation to situation. Consider a very different situation, in which an individual is unprepared for an impromptu work presentation. As he stumbles through the presentation, and his goals shifts to minimizing damage to his professional reputation, the brain mobilizes a different situated conceptualization underlying fear. Predictions generated by the situated conceptualization initiate changes in the brain and body that are specific to the situation. In this situation, for example, visceromotor changes might inhibit the motor system from performing further actions unless absolutely necessary. Shifts in attention occur to organize thoughts involved in a compensatory strategy and to monitor the supervisor's reactions. Perceptual changes occur such that the supervisor's facial actions are interpreted as a frown instead of a neutral expression. As this example illustrates, different situated conceptualizations develop for an emotion concept like fear to initiate changes via prediction and inference as efficiently as possible in the situation.

Similar to other abstract concepts, we assume that goals are central to the situated conceptualizations that underlie emotion concepts. It might be tempting to also assume a one-to-one relationship between goals and emotion concepts (e.g., the goal of avoiding harm and suffering involved in all situated conceptualizations underlying fear). As discussed earlier, however, it appears that goals often vary across the various situated conceptualizations that underlie an emotion concept (e.g., when fear is experienced during thrill seeking), and that goals like avoiding potential harm/distress often operate very differently when grounded in specific situations.

Finally, we assume that the situated conceptualizations underlying emotion concepts are largely assembled based on prior learning, with new learning and updating occurring routinely as novelty is encountered and/or as language is involved in assembling concepts. We further assume that situated conceptualization underlies explicit and implicit forms of emotion regulation, with language often playing a central role in shaping trajectories of situated conceptualization (an extended discussion is beyond the scope of this chapter but see Barrett, Wilson-Mendenhall, & Barsalou, 2014).

Concepts Underlie How Emotions Function

In closing, we would like to briefly address how a situated conceptualization approach built on prediction and inference impacts how we view the functions of emotions, and by doing so, situate our theoretical approach with other theoretical approaches to emotion. Theories of emotions routinely assume that specific emotions or appraisal mechanisms evolved to serve a specific function that is adaptive. Basic emotion models typically assume that each discrete emotion system for fear, anger, disgust, sadness, and so forth evolved to deal with an important kind of problem related to survival, such as escaping predators (Tracy & Randles, 2011). Another major class of theories—appraisal theories—typically suggest that emotions reflect appraisals of the changing environment that function to detect situations relevant to an organism's well-being (Moors, Ellsworth, Scherer, & Frijda, 2013).

A situated conceptualization approach suggests that function is grounded in *learning and mobilizing* situated patterns to guide responding, and this functioning is not specific to emotional experiences. In general, situated approaches to the mind typically view the brain as a coordinated system designed to use information captured during prior situations (and available in memory) to flexibly infer what is currently happening and what to do about it—dynamically shaping moment-to-moment responding in the form of perceiving, coordinating action, regulating the body, and organizing thoughts (Aydede & Robbins, 2009; Barrett, 2013; Barsalou, 2003b, 2009; Glenberg, 1997; Mesquita, Barrett, & Smith, 2010; Wilson-Mendenhall, Barrett, et al., 2013). From a biological systems perspective, it is generally adaptive to use prior experience to predict what will happen in the present or future. This approach is largely consistent with an emerging family of psychological construction theories of emotion (Barrett, 2012; Clore & Ortony, 2013; Cunningham, Dunfield, & Stillman, 2014; Russell, 2003). Situated conceptualizations are especially central in the conceptual act theory

(Barrett, 2006, 2012, 2013), and much of what we have proposed and developed here is inspired by this theoretical approach.

Because a situated conceptualization approach offers a very different perspective on the functions of emotions than has traditionally been discussed, it motivates new research questions. From our perspective, it is critical to understand how situated conceptualizations develop and change. Instead of assuming that specific emotions or appraisals are inherently adaptive, a situated conceptualization approach suggests starting with the basic processes that scaffold attention and learning to establish the conceptual patterns that drive prediction. A possible starting point is that initial goals and motivations are grounded in ensuring immediate and prospective integrity of internal physiology (Critchley & Harrison, 2013), with the increasingly complex conceptual patterns underlying emotions emerging through attention and learning. If we assume that the patterns underlying affect and emotion are learned to a large degree, developmental and lifespan perspectives become increasingly important. Caregivers and close others play a large role in shaping emotional experiences throughout life, with crucial emotional development occurring early in life (Lindquist, MacCormack, & Shablack, 2015).

From our perspective, it is not surprising that emotional experiences vary dramatically within an individual (from situation to situation) and across individuals. Our approach suggests that it is important to understand this variability, and that understanding how situated conceptualizations create this variability could inform how we approach mental health and well-being. Some adults, for example, do not appear to distinguish among discrete emotions like fear, sadness, and anger that are typically unpleasant (Lindquist & Barrett, 2008)—similar to what has been characterized as the initial stages of learning discrete emotions during childhood (Widen & Russell, 2010). Initial evidence suggests that this lack of "granularity" impacts mental health (Demiralp et al., 2012; Suvak et al., 2011; Tugade, Fredrickson, & Barrett, 2004). Restructuring situated conceptualizations and learning new situated conceptualizations (often guided by others) arguably underlies the flexibility to cope with life's challenges. Many of the positive changes that occur in psychotherapy and in supportive relationships could be understood in terms of situated conceptualization (Barrett et al., 2014).

A situated conceptualization approach to emotion differs from many traditional approaches to emotion. From this perspective, the emphasis is on learning, variability, and change, which is how function is understood. Perhaps most importantly, assuming conceptual processing is fundamental to emotion, and thereby changing how we understand the functions of emotion, holds promise for illuminating important individual differences in emotional experiences and vulnerabilities for poor mental health.

Notes

1. Even this may not be that automatic—people learn to sleep through morning alarms, which may become too predictable to be arousing.
2. The neural patterns that underlie initial concepts are developing early in the first year of life (Bergelson & Swingley, 2012; Murphy, 2002; Tincoff & Jusczyk, 1999, 2012) and perhaps before then in the prenatal environment (e.g., patterns that underlie recognizing and responding to a mother's voice; DeCasper & Spence, 1986; Voegtline, Costigan, Pater, & DiPietro, 2013). Because concepts develop to predict what will occur in situations, concepts for experiences that are salient to infants (and that are made salient by parents) will develop first. What is salient to an infant initially may reflect the biological organization that scaffolds learning.
3. In our framework, no strict definitions distinguish affect from emotion. We primarily discuss the categories of emotional experiences that are a focus in the literature (e.g., fear, anger, sadness, happiness), but we assume that many different concepts develop for affective and emotional experiences.
4. Interoceptive activity (involving visceromotor changes) is not typically considered in models of attention, which instead focus on the sense modalities that interface with the external world.
5. For this reason, it is important to distinguish selection of interoceptive activity from "interoceptive awareness," which involves conscious, focal attention to bodily states (Kleckner & Quigley, 2014).
6. Efficiency is not synonymous with well-being. It is generally adaptive to use prior experience to guide responding in the present. Because prediction is experience dependent, however, learned conceptual structures could become entrenched and drive responding in a way that is unhealthy.
7. The term "situated" takes on a broad meaning in our view, referring to the distributed neural activity involved in constructing situations, which reflects the dynamic actions that individuals engage in, and the events, internal bodily sensations, and mentalizing that they experience, as well as the perceptions of the environmental setting and the physical entities

and individuals it contains (Wilson-Mendenhall, Barrett, & Barsalou, 2013; Wilson-Mendenhall et al., 2011).

8. The situated conceptualizations that underlie a concept often exist without any conceptual "core." Instead, situated conceptualizations tend to vary in their similarity to one another, with regularities clustering in various ways, and with no regularities that are common to all situated conceptualizations (e.g., Lakoff, 1987; Rosch & Mervis, 1975; see also Lebois, Wilson-Mendenhall, & Barsalou, 2015). Furthermore, a representation that is separate from individual instances is not required to represent a concept—"prototypes" can be constructed dynamically (Barsalou, 1987; Murphy, 2002).
9. Kopp (1989) discusses the transition from "preadapted programs" to "elemental cognitive" processes. The elemental cognitive mechanism "implicates perceptual discrimination of an event, memory of past experiences in similar situations and learned associations of contingencies, and current self-needs and goal states" (p. 364).
10. These situated, preverbal patterns may underlie infant behaviors that adults would describe as fear (e.g., when abruptly scolded, when meeting "strange" novel people) or anger (e.g., when a caregiver departs, when unable to manipulate a toy). It is possible that these patterns are the starting points for more complex concepts that later become associated with the words "fear" and "anger." It is also possible, however, that these patterns change substantially before children begin to organize them in memory around discrete emotion words (as situated conceptualizations develop through language-based learning).
11. Situated conceptualizations can develop as concepts mobilize to interpret a situation that is presently occurring or to interpret a situation that is occurring as a reconstruction of past experience or a projection of future possibilities. Situated conceptualizations underlying fear, for example, can develop when simulating potential pain or discomfort (e.g., during a car crash) via concepts in memory.

REFERENCES

Akhtar, N., & Tomasello, M. (2000). The social nature of words and word learning. In R. M. Golinkoff, K. Hirsh-Pasek, L. Bloom, L. B. Smith, A. L. Woodward, N. Akhtar, et al. (Eds.), *Becoming a word learner: A debate on lexical acquisition* (pp. 115–135). Oxford, UK: Oxford University Press.

Allen, S. W., & Brooks, L. R. (1991). Specializing the operation of an explicit rule. *Journal of Experimental Psychology: General, 120*(1), 3–19.

Aslin, R. N., & Newport, E. L. (2012). Statistical learning: From acquiring specific items to forming general rules. *Current Directions in Psychological Science, 21*(3), 170–176.

Aydede, H., & Robbins, P. (Eds.). (2009). *The Cambridge handbook of situated cognition*. New York: Cambridge University Press.

Bahrick, L. E., Lickliter, R., & Flom, R. (2004). Intersensory redundancy guides the development of selective attention, perception, and cognition in infancy. *Current Directions in Psychological Science, 13*(3), 99–102.

Bannert, M. M., & Bartels, A. (2013). Decoding the yellow of a gray banana. *Current Biology, 23*(22), 2268–2272.

Bar, M. (2004). Visual objects in context. *Nature Reviews Neuroscience, 5*(8), 617–629.

Barrett, L. F. (2006). Emotions as natural kinds? *Perspectives on Psychological Science, 1*, 28–58.

Barrett, L. F. (2009). Variety is the spice of life: A psychological construction approach to understanding variability in emotion. *Cognition and Emotion, 23*(7), 1284–1306.

Barrett, L. F. (2012). Emotions are real. *Emotion, 12*(3), 413–429.

Barrett, L. F. (2013). Psychological construction: A Darwinian approach to the science of emotion. *Emotion Review, 5*, 379–389.

Barrett, L. F., & Bar, M. (2009). See it with feeling: Affective predictions during object perception. *Philosophical Transactions of the Royal Society of London: Biological Sciences, 364*(1521), 1325–1334.

Barrett, L. F., Lindquist, K. A., Bliss-Moreau, E., Duncan, S., Gendron, M., Mize, J., et al. (2007). Of mice and men: Natural kinds of emotions in the mammalian brain? A response to Panksepp and Izard. *Perspectives on Psychological Science, 2*(3), 297–311.

Barrett, L. F., & Simmons, W. K. (2015). Interoceptive predictions in the brain. *Nature Reviews Neuroscience, 16*, 419–429.

Barrett, L. F., Wilson-Mendenhall, C. D., & Barsalou, L. W. (2014). A psychological construction account of emotion regulation and dysregulation: The role of situated conceptualizations. In J. J. Gross (Ed.), *The handbook of emotion regulation* (2nd ed., pp. 447–465). New York: Guilford Press.

Barrett, L. F., Wilson-Mendenhall, C. D., & Barsalou, L. W. (2015). The conceptual act theory: A road map. In L. F. Barrett & J. A. Russell (Eds.), *The psychological construction of emotion* (pp. 83–110). New York: Guilford Press.

Barsalou, L. W. (1987). The instability of graded structure: Implications for the nature of concepts. In U. Neisser (Ed.), *Concepts and conceptual development: Ecological and intellectual factors in categorization* (pp. 101–140). Cambridge, UK: Cambridge University Press.

Barsalou, L. W. (1991). Deriving categories to achieve goals. In G. H. Bower (Ed.), *The psychology of learning and motivation: Advances in research and theory* (Vol. 27, pp. 1–64). San Diego, CA: Academic Press.

Barsalou, L. W. (1999). Perceptual symbol systems. *Behavioral and Brain Sciences, 22*(4), 577–609.

Barsalou, L. W. (2003a). Abstraction in perceptual symbol systems. *Philosophical Transactions of the Royal Society of London: Biological Sciences, 358*(1435), 1177–1187.

Barsalou, L. W. (2003b). Situated simulation in the human conceptual system. *Language and Cognitive Processes, 18*, 513–562.

Barsalou, L. W. (2005). Abstraction as dynamic interpretation in perceptual symbol systems. In L. Gershkoff-Stowe & D. Rakison (Eds.), *Carnegie Symposium Series. Building object categories* (pp. 389–431). Mahwah, NJ: Erlbaum.

Barsalou, L. W. (2008a). Grounded cognition. *Annual Review of Psychology, 59*, 617–645.

Barsalou, L. W. (2008b). Situating concepts. In P. Robbins & M. Aydede (Eds.), *Cambridge handbook of situated cognition* (pp. 236–263). New York: Cambridge University Press.

Barsalou, L. W. (2009). Simulation, situated conceptualization, and prediction. *Philosophical Transactions of the Royal Society of London: Biological Sciences, 364*(1521), 1281–1289.

Barsalou, L. W. (2011). Integrating Bayesian analysis and mechanistic theories in grounded cognition. *Behavioral and Brain Sciences, 34*, 191–192.

Barsalou, L. W. (2012). The human conceptual system. In M. J. Spivey, K. McRae, & M. Joanisse (Eds.), *The Cambridge handbook of psycholinguistics* (pp. 239–258). New York: Cambridge University Press.

Barsalou, L. W., Breazeal, C., & Smith, L. B. (2007). Cognition as coordinated non-cognition. *Cognitive Processing, 8*(2), 79–91.

Barsalou, L. W., & Hale, C. R. (1993). Components of coceptual representation: From feature lists to recursive frames. In I. Van Mechelen, J. Hampton, R. Michalski, & P. Theuns (Eds.), *Categories and concepts: Theoretical views and inductive data analysis* (pp. 97–144). San Diego, CA: Academic Press.

Barsalou, L. W., Niedenthal, P. M., Barbey, A., & Ruppert, J. (2003). Social embodiment. In B. Ross (Ed.), *The psychology of learning and motivation* (Vol. 43, pp. 43–92). San Diego, CA: Academic Press.

Barsalou, L. W., Santos, A., Simmons, W. K., & Wilson, C. D. (2008). Language and simulation in conceptual processing. In M. De Vega, A. M. Glenberg, & A. Graesser (Eds.), *Symbols, embodiment, and meaning* (pp. 245–283). Oxford, UK: Oxford University Press.

Barsalou, L. W., & Weimer-Hastings, K. (2005). Situating abstract concepts. In D. Pecher & R. Zwann (Eds.), *Grounding cognition: The role of perception and action in memory, language, and thought* (pp. 129–163). New York: Cambridge University Press.

Bastos, A. M., Usrey, W. M., Adams, R. A., Mangun, G. R., Fries, P., & Friston, K. J. (2012). Canonical microcircuits for predictive coding. *Neuron, 76*(4), 695–711.

Benedict, H. (1979). Early lexical development: Comprehension and production. *Journal of Child Language, 6*, 183–200.

Bergelson, E., & Swingley, D. (2012). At 6–9 months, human infants know the meanings of many common nouns. *Proceedings of the National Academy of Sciences of the United States of America, 109*(9), 3253–3258.

Bloom, L. (1993). *The transition from infancy to language*. New York: Cambridge University Press.

Bretherton, I., & Beeghly, M. (1982). Talking about internal states: The acquisition of an explicit theory of mind. *Developmental Psychology, 18*(6), 906–921.

Bretherton, I., Fritz, J., Zahn-Waxler, C., & Ridgeway, D. (1986). Learning to talk about emotions: A functionalist perspective. *Child Development, 57*, 529–548.

Busse, L., Roberts, K. C., Crist, R. E., Weissman, D. H., & Woldorff, M. G. (2005). The spread of attention across modalities and space in a multisensory object. *Proceedings of the National Academy of Sciences of the United States of America, 102*(51), 18751–18756.

Chaigneau, S. E., Barsalou, L. W., & Zamani, M. (2009). Situational information contributes to object categorization and inference. *Acta Psychologica, 130*(1), 81–94.

Chun, M. M., Golomb, J. D., & Turk-Browne, N. B. (2011). A taxonomy of external and internal attention. *Annual Review of Psychology, 62*, 73–101.

Clark, A. (2006). Language, embodiment, and the cognitive niche. *Trends in Cognitive Sciences, 10*(8), 370–374.

Clark, A. (2013). Whatever next?: Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences, 36*(3), 181–204.

Clore, G. L., & Ortony, A. (2013). Psychological construction in the OCC model of emotion. *Emotion Review, 5*(4), 335–343.

Cohen, L. B., & Amsel, G. (1998). Precursors to infants' perception of the causality of a simple event. *Infant Behavior and Development, 21*(4), 713–731.

Cohen, L. B., Chaput, H. H., & Cashon, C. H. (2002). A constructivist model of infant cognition. *Cognitive Development, 17*(3-4), 1323–1343.

Condon, P., Wilson-Mendenhall, C. D., & Barrett, L. F. (2014). What is a positive emotion?: The psychological construction of pleasant fear and unpleasant happiness. In M. Tugade, M. Shiota, & L. Kirby (Eds.), *The handbook of positive emotion* (pp. 60–81). New York: Guilford Press.

Corbetta, M., Patel, G., & Shulman, G. L. (2008). The reorienting system of the human brain: From environment to theory of mind. *Neuron, 58*(3), 306–324.

Critchley, H. D., & Harrison, N. A. (2013). Visceral influences on brain and behavior. *Neuron, 77*(4), 624–638.

Cunningham, W. A., Dunfield, K. A., & Stillman, P. E. (2014). Emotional states from affect dynamics. *Emotion Review, 5*(4), 344–355.

DeCasper, A. J., & Spence, M. J. (1986). Prenatal maternal speech influences newborns' perception of speech sounds. *Infant Behavior and Development, 9*, 133–150.

Demiralp, E., Thompson, R. J., Mata, J., Jaeggi, S. M., Buschkuehl, M., Barrett, L. F., et al. (2012). Feeling blue or turquoise?: Emotional differentiation in major depressive disorder. *Psychological Science, 23*(11), 1410–1416.

Fehr, B., & Russell, J. A. (1984). Concept of emotion viewed from a prototype perspective. *Journal of Experimental Psychology: General, 113*, 464–486.

Fehr, B., & Russell, J. A. (1991). The concept of love viewed from a prototype perspective. *Journal of Personality and Social Psychology, 60*, 425–438.

Feldman, R. (2007). Parent–infant synchrony and the construction of shared timing: Physiological precursors, developmental outcomes, and risk conditions. *Journal of Child Psychology and Psychiatry and Allied Disciplines, 48*(3-4), 329–354.

Feldman, R., & Greenbaum, C. W. (1997). Affect regulation and synchrony in mother–infant play as precursors to the development of symbolic competence. *Infant Mental Health Journal, 18*(1), 4–23.

Friston, K. (2012). Predictive coding, precision and synchrony. *Cognitive Neuroscience, 3*(3-4), 238–239.

Friston, K., & Kiebel, S. (2009). Predictive coding under the free-energy principle. *Philosophical Transactions of the Royal Society of London, Series B: Biological Sciences, 364*(1521), 1211–1221.

Gilhooly, K. J., & Gilhooly, M. L. (1980). The validity of age-of-aquisition ratings. *British Journal of Psychology, 71*, 105–110.

Glenberg, A. M. (1997). What memory is for. *Behavioral and Brain Sciences, 20*(1), 1–19.

Harrison, N. A., Gray, M. A., Gianaros, P. J., & Critchley, H. D. (2010). The embodiment of emotional feelings in the brain. *Journal of Neuroscience, 30*(38), 12878–12884.

Johnson, W. A., & Dark, V. J. (1986). Selective attention. *Annual Review of Psychology, 37*, 43–75.

Kirkham, N. Z., Slemmer, J. A., & Johnson, S. P. (2002). Visual statistical learning in infancy: Evidence for a domain general learning mechanism. *Cognition, 83*(2), B35–B42.

Kleckner, I. R., & Quigley, K. (2014). An approach to mapping the neurophysiological state of the body to affective experience. In L. F. Barrett & J. Russell (Eds.), *The psychological construction of emotion*. New York: Guilford Press.

Klein, R. M. (2000). Inhibition of return. *Trends in Cognitive Sciences, 4*(4), 138–147.

Kopp, C. B. (1989). Regulation of distress and negative emotions: A developmental view. *Developmental Psychology, 25*(3), 243–254.

Kroes, M. C., & Fernandez, G. (2012). Dynamic neural systems enable adaptive, flexible memories. *Neuroscience and Biobehavioral Reviews, 36*(7), 1646–1666.

Kuperman, V., Stadthagen-Gonzalez, H., & Brysbaert, M. (2012). Age-of-acquisition ratings for 30,000 English words. *Behavior Research Methods, 44*(4), 978–990.

Kuppens, P., Van Mechelen, I., Smits, D. J. M., De Boeck, P., & Ceulemans, E. (2007). Individual differences in patterns of appraisal and anger experience. *Cognition and Emotion, 21*(4), 689–713.

Lakoff, G. (1987). *Women, fire, and dangerous things: What categories reveal about the mind*. Chicago: University of Chicago Press.

Lebois, L. A. M., Wilson-Mendenhall, C. D., & Barsalou, L. W. (2015). Are automatic conceptual cores the gold standard of semantic processing?: The context-dependence of spatial meaning in grounded congruency effects. *Cognitive Science, 39*(8), 1764–1801.

Lebois, L. A. M., Wilson-Mendenhall, C. D., Simmons, W. K., Barrett, L. F., & Barsalou, L. W. (2016). *Learning emotions*. Manuscript submitted for publication.

Lindquist, K. A., & Barrett, L. F. (2008). Emotional complexity. In M. Lewis, J. M. Haviland-Jones, & L. F. Barrett (Eds.), *The handbook of emotion* (pp. 513–530). New York: Guilford Press.

Lindquist, K. A., MacCormack, J. K., & Shablack, H. (2015). The role of language in emotion: Predictions from psychological constructionism. *Frontiers in Psychology, 6*, 444.

Lindquist, K. A., Satpute, A. B., & Gendron, M. (2015). Does language do more than communicate emotion? *Current Directions in Psychological Science, 24*, 99–108.

Lindquist, K. A., Siegel, E. H., Quigley, K. S., & Barrett, L. F. (2013). The hundred-year emotion war: Are emotions natural kinds or psychological constructions? Comment on Lench, Flores, and Bench (2011). *Psychological Bulletin, 139*(1), 255–263.

Lindquist, K. A., Wager, T. D., Kober, H., Bliss-Moreau, E., & Barrett, L. F. (2012). The brain basis of emotion: A meta-analytic review. *Behavioral and Brain Sciences, 35*(3), 121–143.

Littenberg, R., Tulkin, S. R., & Kagan, J. (1971). Cognitive components of separation anxiety. *Developmental Psychology, 4*(3), 387–388.

Logan, G. D. (1994). Spatial attention and the apprehension of spatial relations. *Journal of Experimental Psychology—Human Perception and Performance, 20*(5), 1015–1036.

McCauley, R. N., & Henrich, J. (2006). Susceptibility to the Muller–Lyer illusion, theory-neutral observation, and the diachronic penetrability of the visual input system. *Philosophical Psychology, 19*(1), 79–101.

Medin, D. L., & Schaffer, M. M. (1978). Context theory of classification learning. *Psychological Review, 85*(3), 207–238.

Mesquita, B., Barrett, L. F., & Smith, E. (Eds.). (2010). *The mind in context*. New York: Guilford Press.

Moll, J., de Oliveira-Souza, R., Moll, F. T., Ignacio, F. A., Bramati, I. E., Caparelli-Daquer, E. M., et al. (2005). The moral affiliations of disgust: A functional MRI study. *Cognitive and Behavioral Neurology, 18*(1), 68–78.

Moors, A., Ellsworth, P. C., Scherer, K. R., & Frijda, N. H. (2013). Appraisal theories of emotion: State of the art and future development. *Emotion Review, 5*(2), 119–124.

Murphy, G. L. (2002). *The big book of concepts*. Cambridge, MA: MIT Press.

Oosterwijk, S., Mackey, S., Wilson-Mendenhall, C., Winkielman, P., & Paulus, M. P. (2015). Concepts in context: Processing mental state concepts with internal or external focus involves different neural systems. *Social Neuroscience, 10*(3), 294–307.

Ortony, A., & Turner, T. J. (1990). What's basic about basic emotions? *Psychological Review, 97*(3), 315–331.

Papies, E. K., & Barsalou, L. W. (2015). Grounding desire and motivated behavior: A theoretical framework and review of empirical evidence. In W. Hofmann & L. F. Nordgren (Eds.), *The psychology of desire* (pp. 36–60). New York: Guildford Press.

Peelen, M. V., & Kastner, S. (2014). Attention in the real world: Toward understanding its neural basis. *Trends in Cognitive Sciences, 18*(5), 242–250.

Petersen, S., Schroijen, M., Molders, C., Zenker, S., & Van den Bergh, O. (2014). Categorical interoception: Perceptual organization of sensations from inside. *Psychological Science, 25*(5), 1059–1066.

Plude, D. J., Enns, J. T., & Brodeur, D. (1994). The development of selective attention: A life-span overview. *Acta Psychologica, 86*, 227–272.

Quigley, K. S., & Barrett, L. F. (2014). Is there consistency and specificity of autonomic changes during emotional episodes?: Guidance from the conceptual act theory and psychophysiology. *Biological Psychology, 98*, 82–94.

Raichle, M. E. (2010). Two views of brain function. *Trends in Cognitive Sciences, 14*(4), 180–190.

Ridgeway, D., Waters, E., & Kuczaj, S. A. (1985). Acquisition of emotion–descriptive language: Receptive and productive vocabulary norms for ages 18 months to 6 years. *Developmental Psychology, 21*(5), 901–908.

Rosch, E. H., & Mervis, C. B. (1975). Family resemblances: Studies in the internal structure of categories. *Journal of Experimental Psychology: General, 7*, 573–605.

Russell, J. A. (1991). In defense of a prototype approach to emotion concepts. *Journal of Personality and Social Psychology, 60*, 37–47.

Russell, J. A. (2003). Core affect and the psychological construction of emotion. *Psychological Review, 110*(1), 145–172.

Russell, J. A., & Fehr, B. (1994). Fuzzy concepts in a fuzzy hierarchy: Varieties of anger. *Journal of Personality and Social Psychology, 67*(2), 186–205.

Saffran, J. R., Aslin, R. N., & Newport, E. L. (1996). Statistical learning by 8-month-old infants. *Science, 274*(5294), 1926–1928.

Schmid, H.-J. (2000). *English abstract nouns as conceptual shells: From corpus to cognition*. Berlin: Mouton de Gruyter.

Schyns, P. G., Goldstone, R. L., & Thibaut, J. P. (1998). The development of features in object concepts. *Behavioral and Brain Sciences, 21*(1), 1–17, discussion 17–54.

Seth, A. K., & Critchley, H. D. (2013). Extending predictive processing to the body: Emotion as interoceptive inference. *Behavioral and Brain Sciences, 36*(3), 227–228.

Shipp, S., Adams, R. A., & Friston, K. J. (2013). Reflections on agranular architecture: Predictive coding in the motor cortex. *Trends in Neurosciences, 36*(12), 706–716.

Smith, L. B., Suanda, S. H., & Yu, C. (2014). The unrealized promise of infant statistical word–referent learning. *Trends in Cognitive Sciences, 18*(5), 251–258.

Smith, L. B., & Yu, C. (2008). Infants rapidly learn word–referent mappings via cross-situational statistics. *Cognition, 106*(3), 1558–1568.

Suvak, M. K., Litz, B. T., Sloan, D. M., Zanarini, M. C., Barrett, L. F., & Hofmann, S. G. (2011). Emotional granularity and borderline personality disorder. *Journal of Abnormal Psychology, 120*(2), 414–426.

Tincoff, R., & Jusczyk, P. W. (1999). Some beginnings of word comprehension in 6-month-olds. *Psychological Science, 10*(2), 172–175.

Tincoff, R., & Jusczyk, P. W. (2012). Six-month-olds comprehend words that refer to parts of the body. *Infancy, 17*(4), 432–444.

Tomasello, M. (1992). The social bases of language aquisition. *Social Development, 1*(1), 67–87.

Tracy, J. L., & Randles, D. (2011). Four models of basic emotions: A review of Ekman and Cordaro, Izard, Levenson, and Panksepp and Watt. *Emotion Review, 3*(4), 397–405.

Treisman, A. M. (1969). Strategies and models of selective attention. *Psychological Review, 76*(3), 282–299.

Treisman, A. M. (1998). Feature binding, attention and object perception. *Philosophical Transactions of the Royal Society of London, Series B: Biological Sciences, 353*(1373), 1295–1306.

Tugade, M. M., Fredrickson, B. L., & Barrett, L. F. (2004). Psychological resilience and positive emotional granularity: Examining the benefits of positive emotions on coping and health. *Journal of Personality, 72*(6), 1161–1190.

Turk-Browne, N. B. (2012). Statistical learning and its consequences. *Nebraska Symposium on Motivation, 59*, 117–146.

Vetter, P., & Newen, A. (2014). Varieties of cognitive penetration in visual perception. *Consciousness and Cognition, 27*, 62–75.

Vigliocco, G., Meteyard, L., Andrews, M., & Kousta, S. (2009). Toward a theory of semantic representation. *Language and Cognition, 1*(2), 219–247.

Voegtline, K. M., Costigan, K. A., Pater, H. A., & DiPietro, J. A. (2013). Near-term fetal response to maternal spoken voice. *Infant Behavior and Development, 36*(4), 526–533.

von dem Hagen, E. A., Beaver, J. D., Ewbank, M. P., Keane, J., Passamonti, L., Lawrence, A. D., et al. (2009). Leaving a bad taste in your mouth but not in my insula. *Social Cognitive and Affective Neuroscience, 4*(4), 379–386.

Wellman, H. M., Harris, P. L., Banerjee, M., & Sinclair, A. (1995). Early understanding of emotion: Evidence from natural language. *Cognition and Emotion, 9*(2/3), 117–149.

Widen, S. C., & Russell, J. A. (2008). Children acquire emotion categories gradually. *Cognitive Development, 23*, 291–312.

Widen, S. C., & Russell, J. A. (2010). Differentiation in preschooler's categories of emotion. *Emotion, 10*(5), 651–661.

Wilson-Mendenhall, C. D., Barrett, L. F., & Barsalou, L. W. (2013). Situating emotional experience. *Frontiers in Human Neuroscience, 7*, 764.

Wilson-Mendenhall, C. D., Barrett, L. F., & Barsalou, L. W. (2015). Variety in emotional life: Within-category typicality of emotional experiences is associated with neural activity in large-scale brain networks. *Social Cognitive and Affective Neuroscience, 10*(1), 62–71.

Wilson-Mendenhall, C. D., Barrett, L. F., Simmons, W. K., & Barsalou, L. W. (2011). Grounding emotion in situated conceptualization. *Neuropsychologia, 49*(5), 1105–1127.

Wilson-Mendenhall, C. D., Simmons, W. K., Martin, A., & Barsalou, L. W. (2013). Contextual processing of abstract concepts reveals neural representations of nonlinguistic semantic content. *Journal of Cognitive Neuroscience, 25*(6), 920–935.

Wu, L. L., & Barsalou, L. W. (2009). Perceptual simulation in conceptual combination: Evidence from property generation. *Acta Psychologica, 132*(2), 173–189.

Yeh, W., & Barsalou, L. W. (2006). The situated nature of concepts. *American Journal of Psychology, 119*(3), 349–384.

Yoshida, H., & Smith, L. B. (2005). Linguistic cues enhance the learning of perceptual cues. *Psychological Science, 16*(2), 90–95.

Zwaan, R. A., & Madden, C. J. (2004). Updating situation models. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 30*(1), 283–288.

Zwaan, R. A., & Radvansky, G. A. (1998). Situation models in language comprehension and memory. *Psychological Bulletin, 123*(2), 162–185.